

Machine Learning Homework 2

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Solution

Suppose $\hat{x}$ is currently at H_i and moved to H_j , then the class i is updated as follows,

$$\begin{aligned}\mathbf{m}_i^* &= \frac{n_i \mathbf{m}_i - \hat{\mathbf{x}}}{n_i - 1} = \mathbf{m}_i + \frac{n_i \mathbf{m}_i - \hat{\mathbf{x}}}{n_i - 1} - \mathbf{m}_i \\ &= \mathbf{m}_i - \frac{\hat{\mathbf{x}} - \mathbf{m}_i}{n_i - 1} \\ J_i^* &= \sum_{\mathbf{x} \in H_i} \|\mathbf{x} - \mathbf{m}_i^*\|^2 - \|\hat{\mathbf{x}} - \mathbf{m}_i^*\|^2 \\ &= \sum_{\mathbf{x} \in H_i} \left\| \mathbf{x} - \mathbf{m}_i + \frac{\hat{\mathbf{x}} - \mathbf{m}_i}{n_i - 1} \right\|^2 - \left\| \frac{n_i}{n_i - 1} (\hat{\mathbf{x}} - \mathbf{m}_i) \right\|^2 \\ &= \sum_{\mathbf{x} \in H_i} \|\mathbf{x} - \mathbf{m}_i\|^2 + \frac{2}{n_i - 1} \sum_{\mathbf{x} \in H_i} (\hat{\mathbf{x}} - \mathbf{m}_i)^T (\mathbf{x} - \mathbf{m}_i) \\ &\quad + \frac{1}{(n_i - 1)^2} \sum_{\mathbf{x} \in H_i} \|\hat{\mathbf{x}} - \mathbf{m}_i\|^2 - \frac{n_i^2}{(n_i - 1)^2} \|\hat{\mathbf{x}} - \mathbf{m}_i\|^2 \\ &= J_i + \frac{n_i}{(n_i - 1)^2} \|\hat{\mathbf{x}} - \mathbf{m}_i\|^2 - \frac{n_i^2}{(n_i - 1)^2} \|\hat{\mathbf{x}} - \mathbf{m}_i\|^2 \\ &\Leftrightarrow J_i^* = J_i - \frac{n_i}{n_i - 1} \|\hat{\mathbf{x}} - \mathbf{m}_i\|^2\end{aligned}$$

According to the update rule, J_i can either remain or decrease.